

INTERNET - Two or more than two computers connecting each other for sharing data and resources is known as network. The internet is a network of networks that makes information easily accessible even different type of computers with different capabilities.

Internet is the system architecture that has method of communication and different technologies by applying these methods we connect various computers worldwide. Sometimes it is referred to as a network of networks that consist of private, public, academics, business and government network of local to global spoke link by array of electronics wireless and optical networking technologies.

The internet emerged in the United States in 1970's but did not become popular to the general public until the early 1990's. It supports human communication by social media, email, chat group, news group and audio video transmission. It allows people to work collaboratively at many different locations. It supports access to the digital information by many applications including world wide network.

Internet firstly researched by US Agencies has ARPANET which means Advance Research Project Agencies of Network.

Internet of Things (IOT) is a network of physical object emb with sensors software and communication technologies enabling them to connect to the internet and exchange data automatically without direct human involvement.

- 1) Focuses on everyday physical object such as devices, machines and systems to computer network.
- 2) Enables real time data collection mon and automated actions. Play keyrole in modern computer networks by extending internet connectivity beyond traditional computer.
- 3) Improves efficiency, accuracy and convince in daily life and industrial application.

Sensors collect data and send it via the network to edge or cloud systems where is data is processed and analyzed result are

	Actuators	Interfaces
Environment	Micro-com- puters Sensors	Mobile, Tablet Laptop, Computer



World Wide Web

Data Management
Data Repositories
Software Technology
Web Application.

delivered to through user through apps or used automatically drive actuators and systems.

IOT Components :- An Iot system is built using multiple inter-connected component that work together to sense the data. Processing, transmitt over network and present useful information to user or trigger actions automatically.

(1) IoT Devices - Physical objects such as machine, an variable that sense data from the

environment and

Sensors and Actuators - Sensors collect environmental data like temperature, motion and while actuator converts digital data into physical actions.

Embedded Processing Devices -

Micro-controllers are micro-processors that process sensor data execute control logic and manage device operations.

Connectivity modules - Enable devices to communicate over networks using technologies such as wifi, bluetooth, cellular networks or LPWAN.

Aggregate

IOT Gateways - Act as data from multiple devices to perform protocol translation and securely transmit data to cloud or the platform.

Cloud and Edge Computing -

Cloud platforms store and analyze large IOT data via edge computing.

process data near the source to reduce latency.

User Interface and Application -

Dashboards, Mobile apps, or web interfaces that allow users to monitor data, control devices and receive alerts.

IOT Architecture -

Application layer

Smart Application

Processing / middle
ware layer

Processing
Information

Transport /
Network layer

Data
Transmission

Sensing layer

Data
Gathering

IOT system are divide into layers to ensure smooth interaction between sensor networks, data processing and application.

1) **Perception (Sensing Layer)** -

Collect real world data, such as

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temperature, humidity, motion, light, or pressure using sensors.

- Uses actuator to perform physical actions like switching devices on/off
- identify and tracks physical object using technology RFID and NFC.

Transport or Network Layer-

- i) Transfers sense data from IoT devices to gateway edge, nodes or cloud
- ii) maintains reliable communication using wire and wireless technologies like wifi, bluetooth, ~~wifi~~ and cellular networks zigbee

Supports communication protocols to ensure efficient and secure data exchange between devices and sys

Processing and Middle ware layer-

- i) Filters, analysis and stores incoming data for further use and decision making

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- ii) Manages connected devices handles inter-operability and supports scalability across large IoT deployments

Application layer-

- ii) Provides dashboards mobile apps or web interface for monitoring and controlling IoT systems.

Presents process data in a meaningful form to supports

Delivers domain specific solutions such as smartphones, ~~for~~ health care monitoring Industrial automation and smart-cities.

Types of Sensors:-

- 1) Image Sensor - Capture visual data from camera and computer vision application.

Temperature Sensor - Measure heat or temperature or changes in environment or object

Gyrosensor (Gyroscope) - Detects angular velocity and orientation of objects

Obstacle Sensor - Identify the presence of obstacle to avoid collisions

RF Sensors - Use radio frequency signals for detection tracking and communication.

IR Sensors - Infrared sensor detect heat signature or motion using infrared radiation.

MQ-02/MQ-05 Gas Sensor -

Sense the presence and concentration of gas such as carbon monoxide, methane, or smoke.

LDR Sensors (Light Dependent Resistor) -

Measures light intensity and brightness levels.

Ultrasonic Distance Sensor -

Calculate distance by emitting ultrasonic waves and measuring their reflection time.

Technologies enabling with IOT:-

RF-ID and NFC - Radio frequency

identification and near field communication are used for automatic identification and short range data exchange. These technologies tell IOT system recognise objects and enables contactless communication.

Embedded system - Embedded systems are dedicated computing tools units integrated into IOT devices. They control sensors and actuators, process collected data and manage communication with other devices on platforms.

Wireless Sensors Network (WSN) -

Wireless sensors network consist of multiple sensor nodes that communicate wirelessly to collect and transmit data. They are widely used for environmental and industrial conditions.

Cloud computing and Big Data:-

Cloud computing provides scalable storage and computing resources for handling large volumes of IOT data via big data technologies support efficient data analysis and management.

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Edge and fog computing process data closer to the source devices rather than sending all data to the cloud. This reduces latency, improves response time and lowers network bandwidth usage.

Growth of Internet - The primary process network or advanced research project agency network (ARPA NET) initially served as a backbone for interconnection of regional academic and military network in the 1970's.

The funding of National Science Foundation Network as a new back-bone in 1980's. As well as private funding of other commercial expansion. Let to lead world wide web participation in the development of new networking technology and the merger of many new network.

The linking of commercial network and enterprises by the early 1990's mark the beginning of transition to the exponential growth as generation of institutional, personal and mobile computer were connected to the network.

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Owner of Internet - No single person or company or government owns the internet fully. The internet is more on a computer concept, done by a physical entity. No person has a copyright over the internet. Parts of internet are owned by countless individuals, companies and government agencies who regulate and control the internet.

(Encryption) $\xrightarrow{\text{Message}}$ (Decryption)
 Source $\rightarrow$ Receiver
 Regulation N/A

The continuous flow of internet is controlled by network administrators. The internet is largely uncontrolled and self-regulated. There is no single centralized organisation that controls the internet. The design of the internet infrastructure makes it hard to regulate. Various governments have tried to regulate the internet for various reasons, usually related to illegal or harmful content on the internet. In this way, different governments regulate the internet by laws.



Anatomy of Internet -

Internet in different groups of computers link by cables and satellite and not control by any single authority but all operating system follow the common network protocol (set of rules).

The term internet includes not the hardware and the software that enables the computers to communicate with each other. When the information is sent across the internet and transmission control protocol (TCP) is used -

- i) This protocol first breaks the information into 2 parts first part is divided into packets which is sent to the client with internet. The second part the client computer send the packet to the local network internet service provider (ISP) from here the packet travel to the many levels of network computer and communication lines until they reach to the final destination.

Net Etiquette - Etiquette is code of polite behaviour in a society. Net Etiquette is a code

of good behaviour on the internet such as email, social media, online chat, web forums, website comment, multiplayer gaming and other type of online communication.

While there is no official list of net etiquettes stick the general idea is to respect others online. Some of the basic net etiquette are as follows -

- 1) Remember the human -

What frequently gets into trouble when communicating with real people there is a human behind the words you are reading. It is easier to respond with negative comment because you are looking a text not on screen rather looking at someone in the eyes.

- 2) Know where you are in cyber space -

Knowing where you are in cyber space with your audience is essential because online online forum

what is good for girl group an not

3) Share expert knowledge - true

One of the main benefits to expand online comm is the ability to share and retrieve knowledge and if you are an expert and have research on share some news so that this is the one of best uses of internet

4) Respect one and other's privacy -

An important net etiquette to is respecting the privacy of the other and we should not publicly identify and post private information about someone specially in form of punishment and revenge

5) Avoid Flames - A Flame is a message that contains strong personal critic comments an attack. Users are advise do not to engage in flaming or participate in flame of wars

OSI Model (Open Structure Interconnection)

The OSI model is conceptual frame work created by International Organization for standardization

Sender

Application Layer

Presentation Layer

Session Layer

Transport Layer

TCP/UDP

Network Layer

Routers & switches

Data Link Layer

Switches & Bridge

Physical Layer

Hub, repeater, Gateway

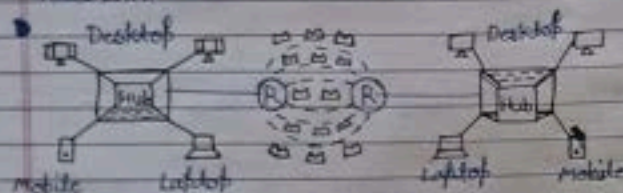
Receiver

to determine describe how data is transmitted across a network using a structured 7 layer architecture. These function are as -

- 1) Divides network communication into 7 functional layer
- 2) Assign specific responsibility to each layer.
- 3) Promotes compatibility between networking system.
- 4) Simplifies network design fundamental implementation and trouble shooting

Routers - Router is a network devices which works as a traffic controller. A main work of router is to chose a congestion free path (Is a route in traffic of on computer networks) that avoids heavy blockages and delays by using real time data to steer data packets.) routing can be of 2 types-

- i) **Static Routing** - Configured manually suitable for small and stable networks.
- ii) **Dynamic Routing** - Automatically updated based on network activity - by ideal for large or changing network.



3) Routers use both LAN and WAN network.

Advantages-

- 1) It provide connection between 2 dissimilar type of network.

2) Transmission rate is very high.

3) It internally uses some algorithm to find out congestion free path

4) It provide both wire and wireless facility

Disadvantages-

1) Router is more expensive compare to other network device

2) Routers are complex to maintain.

3) Security issue.

4) It only work with routed/switchable protocol.

Hub - Hub is a network device that is used to connect multiple computers or device in a network within a LAN.

1) All the information send to the hub is automatically send to each node to every device.

2) Hub is less expensive, less intelligent, less complicated.

3) Hub generally used to connect computer.

In a lab.

4) Transmission mode of hub is half duplex

5) It receives data packets on one-bound port and dynamic broadcast to them to all other ports

Advantages-

- 1) The Hub can broadcast the message.
- 2) It is less expensive that anybody can use it.
- 3) Easy installation.
- 4) Robust (so strong)

Disadvantages-

- 1) If the hub is failed the entire network will failed.
- 2) We can't send personal or private data through hub.
- 3) Hub does not provide any security
- 4) Hub can't support full duplex

5) transmission mode.

Repeaters - Repeater is a network device through which we can boost up the weak signal when the signal travel in the network after travelling some distance the intensity of the network signal becomes low. It is used in wired and wireless devices.

Advantages-

- 1) It is used to regenerate the weak signals.
- 2) It is cheaper than other network device.
- 3) Repeater has the ability to change legth length of the signal.
- 4) Increase or maintain the signal performance

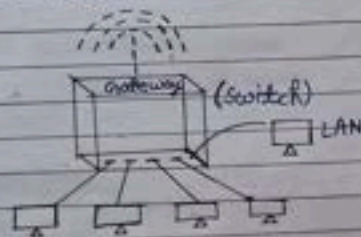
Disadvantages-

- 1) It required no. of repeaters after some distance.
- 2) Repeater also unable to connect

dis-similar type of network

3) They can't reduce network traffic.

Gateway - Gateway is a hardware device i.e. used to connect two dis-similar type of network. It allow us to send and receive data through internet even if is LAN network. It operates all seven layers of OSI Model.



Assignment - 2

Advantages -

- 1) It connects two network device which have different protocol.
- 2) It operate all 7 layers of protocol.
- 3) We can't access the internet without a gateway.
- 4) It provide some security.

Disadvantages -

- 1) It is more expensive.
- 2) Data transmission rate is very slow.
- 3) Difficult to maintain as well as very complex.
- 4) It is less intelligent.

Switch - A switch is a high speed networking device that connects devices (computer, printers, servers within a LAN).

Unlike hub switch learns the mac address of every connected devices.

It sends data only to the specific port where the destination device is connected

Reduces collisions and improves security

Advantages → 1) Fast Performance -

- 1) Directs data only to the intended devices, eliminating traffic jams
- 2) Full Duplex - Devices can send and receive data at the same time

Disadvantages →

- 1) It is more expensive than basic network.
- 2) If a switch breaks, all connected devices lose communication.

Bridge - Bridge connects two separate LAN segments to make them appear as one.

- 2) Filters traffic based on mac addresses to keep local traffic control reducing congestion.

Largely replaced by switches (which are essentially multiport bridge)

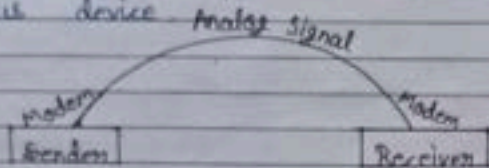
Access Point - It is a networking hardware device that allows wifi enabled devices such as laptops and smartphones to connect to a wired network.

- 2) Acts as a bridge between ethernet and wireless wifi devices
- 3) An AP only provides wifi signals and it does not moved traffic on assign IP addresses

Modem - A Modem is a hardware device that connect a computer and route it to a broadband network. It is a short-term form of modulation and demodulation. A modem is a hardware device that bridges the gap between digital computer data and analog signals. Allowing devices to connect to the internet via telephone cable or fibre lines. Also used to send and receive information over all telephone lines. When sending a signal the device convert digital data into analog they need transmit it over a network telephone line similarly when an analog

signal receive by the modem it converts it back to the digital signal.

Modem are refers to the asynchronous device.



Internet Governance -

IG is a development and application of shared principle, norms, regulations, and decision making procedures and program that save the evolution and use of Internet.

It covers activity such as development and coordination of technical standards, operations of critical infrastructure and public policy issue conceptually internet governance into following layers as -

- 1) Physical Infrastructure Layer
- 2) Code / Logical Layer
- 3) Content Layer
- 4) Security Layer

Currently organisation such as internet corporation of assigned names and

members. It is an organisation which is incorporated in the real state of California government by U.S. laws and accountable to the U.S. Department of Commerce under an agreement to performing internet assigned no. of authorities and is responsible for technical operations of root and domain name infrastructure.

Internet Engineering Task Force (IETF) is responsible for developing protocols and play important roles.

Impact of Internet on Society +

- # The internet has a large impact on our society and on the way of living.
- # In some cases using the internet can save a lot of money and time. Shopping on the internet allow for instant price comparisons without leaving home.
- # Relationship have changed based on internet access. E-mail chat and instant message.
- # Education have expanded to allow people to get full degree online without leaving the comfort of home.
- # People need and get to know about each other and can remain in closer contact with friends and family and

help to keep everyone inform on the event.

E-commerce → It is digital platform for buying and selling of goods and services or transferring products. It make business and consumer globally connected by securing payment from

E-commerce is buying and selling of goods and services on transferring the funds and data over the electronic network.

There are different types of e-commerce services which are consumed by the user as always as merchant. Some of them are follows-

- i) Business to Business
- ii) Business to Consumer
- iii) Consumer to Business
- iv) Consumer to Consumer

The term e-commerce is also known as E-business. The term E-business also used as E-bay that is used in reference to the traditional transaction process that make up online retail shopping.

Key Features →

- 1) Product Catalogue
- 2) Shopping Cart
- 3) Checkout and payment processing
- 4) Shipping and order management
- 5) Customer accounts and loyalty program
- 6) Marketing and Analytics
- 7) Mobile optimization.

Business to Business - B2B e-commerce refers to electronic exchange of products, services or information between businesses rather than between business and consumer. For example Amazon Business for corporate bulk purchasing, Ali Baba for global wholesale service, IndiaMart, Udaan etc.

Business to Consumer - B2C is the retail part of e-commerce on the internet. It is used when business sell product services or information directly to consumer. For example Online retail stores, Digital streaming services/subscriptions, multi vendors market place.

Consumer to Business - C2B is a type of e-commerce in which consumer make products and services, then online companies.

for ex- User generated content

Consumer to Consumer -

C2C is a type of e-commerce in which consumer to consumer exchange product, services, data with each other online for ex- Quicken, etc.